

Objective Mackey and Tambara functors via parametrized categories

Ross Street
CoACT
Macquarie University
Dharug Wallumattagal Country

Australian Category Seminar
Macquarie University

Some take-home points

- ▶ Lindner (1976): Mackey functors as pairs of functors are equivalently single functors with domain a category of spans.
- ▶ Tambara (1993): TraceNormRestriction-functors are equivalently certain functors out of a category of polynomials.
- ▶ Here: Objective Mackey and Tambara functors are parametrized categories which have local finite products and satisfy some parametrized completeness and cocompleteness restriction.
- ▶ Equivalently the original parametrizing base can be replaced, for objective Mackey functors by a bicategory of spans, while for objective Tambara functors by a bicategory obtained by iterating the span construction; these iterated spans are polynomials.
- ▶ There is a distributive law relating objective Mackey functors to objective Tambara functors analogous to the distributive law relating abelian groups to commutative rings.
- ▶ Relevant is the 2-category Cat_+ of categories admitting finite coproducts, both as a closed base and as a skew-closed base.

Definition

A *protocalibration* of a bicategory $\mathcal{C}$ is a set $\mathcal{R}$ of morphisms when it satisfies the four conditions:

- I. all equivalences are in $\mathcal{R}$ and any morphism isomorphic to one in $\mathcal{R}$ is in $\mathcal{R}$;
- C. composites of morphisms in $\mathcal{R}$ are in $\mathcal{R}$;
- P. for all $U \xrightarrow{f} W$ in $\mathcal{R}$ and all $V \xrightarrow{g} W$ in $\mathcal{C}$, the bicategorical pullback $P \xrightarrow{\bar{f}} V$ of f along g exists and is in $\mathcal{R}$;
- G. every morphism in $\mathcal{R}$ is a groupoid opfibration.

The set $\mathcal{P}$ of powerful morphisms is a protocalibration of any category $\mathcal{C}$ with pullbacks.

$\mathcal{R}$ -cocompletion

Suppose $\mathcal{R}$ is a protocolibration of the bicategory $\mathcal{C}$.

The $\mathcal{R}$ -cocompletion $\overrightarrow{\mathbb{X}}$ of $\mathbb{X} \in \text{CAT}\mathcal{C} = \text{Ps}(\mathcal{C}^{\text{op}}, \text{Cat})$ is as follows:

An object of the category $\overrightarrow{\mathbb{X}}U$ is a pair $(U \xleftarrow{u} S, x \in \mathbb{X}S)$ with $u \in \mathcal{R}$; we might think of such an object as a U -indexed family of generalised objects of $\mathbb{X}$. A morphism

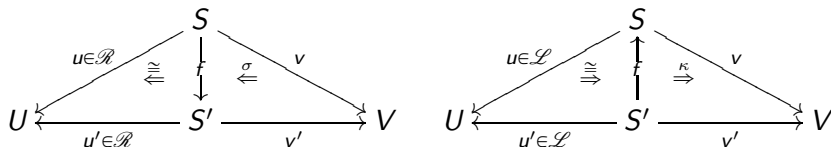
$$(U \xleftarrow{u} S, x \in \mathbb{X}S) \rightarrow (U \xleftarrow{u'} S', x' \in \mathbb{X}S')$$

of $\overrightarrow{\mathbb{X}}U$ is an isomorphism class of triples $(S \xrightarrow{w} S', u'w \cong u, x \xrightarrow{\xi} (\mathbb{X}w)x')$ where ξ is a morphism of $\mathbb{X}S$; if we identify representables with their representing objects, the picture of such a morphism is a lax morphism of spans in $\text{CAT}\mathcal{C}$.

$$\begin{array}{ccccc}
 & & S & & \\
 & \swarrow^{u \in \mathcal{R}} & \downarrow w & \searrow^x & \\
 U & \xleftarrow{u' \in \mathcal{R}} & S' & \xrightarrow{x'} & \mathbb{X}
 \end{array}
 \quad
 \begin{array}{c}
 \cong \\
 \xi \\
 \cong
 \end{array}$$

$\mathcal{R}$ -Spans and $\mathcal{R}$ -cocompleteness

Let $\mathcal{R}$ be a protocalibration of the bicategory $\mathcal{C}$. A (left) $\mathcal{R}$ -span from U to V in $\mathcal{C}$ is a diagram $U \xleftarrow{u} S \xrightarrow{v} V$ in $\mathcal{C}$ with $u \in \mathcal{R}$. There is a bicategory $\mathrm{Spn}_{\mathcal{R}}\mathcal{C}$ whose objects are those of $\mathcal{C}$, whose morphisms are the $\mathcal{R}$ -spans, and whose 2-morphisms (denoted $[f, \sigma]$ by abuse of language) are the isomorphism classes of diagrams on the left below.



Let $\mathcal{L}$ be a protocalibration of the bicategory $\mathcal{C}^{\mathrm{co}}$. Put

$${}_{\mathcal{L}}\mathrm{Spn}\mathcal{C} = (\mathrm{Spn}_{\mathcal{L}}\mathcal{C}^{\mathrm{co}})^{\mathrm{co}}.$$

See on the right in diagram above.

Example

Let $\mathcal{E}$ be a category with pullbacks and every morphism powerful. The groupoid fibrations (= left-adjoint morphisms) in the bicategory $\mathrm{Spn}\mathcal{E}$ of spans in $\mathcal{E}$ is a protocolibration of $(\mathrm{Spn}\mathcal{E})^{\mathrm{co}}$.

Proposition

Restriction along the opposite of

$$(-)_* : \mathcal{C} \rightarrow \mathrm{Spn}_{\mathcal{R}} \mathcal{C}$$

defines a bicategorical equivalence

$$\mathrm{CAT}(\mathrm{Spn}_{\mathcal{R}} \mathcal{C}) \sim \mathrm{CAT}_{\mathcal{R}} \mathcal{C} .$$

Dually,

$$\mathrm{CAT}({}_{\mathcal{L}} \mathrm{Spn} \mathcal{C}) \sim {}_{\mathcal{L}} \mathrm{CAT} \mathcal{C} .$$

Compatible protocalibrations

Definition

Suppose $\mathcal{L}$ and $\mathcal{R}$ are protocalibrations of the category $\mathcal{E}$. We say the pair $(\mathcal{L}, \mathcal{R})$ is *compatible* when $\mathcal{L} \subseteq \mathcal{P} \cap \mathcal{R}$ and $\Pi_r v \in \mathcal{R}$ whenever $r \in \mathcal{L}$ and $v \in \mathcal{R}$.

Example

In a category $\mathcal{E}$ with pullbacks, the maximum example of a compatible pair is $(\mathcal{P}, \mathcal{E})$.

Lemma

Suppose the pair $(\mathcal{L}, \mathcal{R})$ of protocalibrations of $\mathcal{E}$ is compatible. Spans in $\mathcal{E}$ with left leg invertible and right leg in $\mathcal{R}$ are morphisms in $\mathcal{L}\text{Spn}\mathcal{E}$ and as such form a protocalibration of $\mathcal{L}\text{Spn}\mathcal{E}$ denoted by $\mathcal{R}_$.*

Suppose $(\mathcal{L}, \mathcal{R})$ is compatible. $\mathcal{L}$ -completion is denoted by $\overleftarrow{\mathbb{X}}$.
There is a bicategorical distributive law

$$\lambda : \overleftarrow{(-)} \circ \overrightarrow{(-)} \longrightarrow \overrightarrow{(-)} \circ \overleftarrow{(-)}$$

where the component functor

$$\lambda_U : \overleftrightarrow{\mathbb{X}} U \longrightarrow \overleftrightarrow{\mathbb{X}} U$$

takes $(U \xleftarrow{u} S, S \xleftarrow{a} A, x \in \mathbb{X}A)$, where $u \in \mathcal{L}$ and $a \in \mathcal{R}$, to

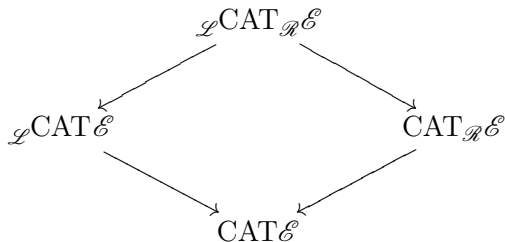
$$(U \xleftarrow{\Pi_u a} B, B \xleftarrow{\bar{u}} P, (\mathbb{X}e)_x \in \mathbb{X}P)$$

where

$$\begin{array}{ccccc} A & \xleftarrow{e} & P & \xrightarrow{\bar{u}} & B \\ & \searrow a & \downarrow \Delta_u \Pi_u a & & \downarrow \Pi_u a \\ & & S & \xrightarrow{u} & U \end{array}$$

The pseudo-algebras for the composite pseudomonad $\mathbb{X} \mapsto \overleftrightarrow{\mathbb{X}}$ on $\text{CAT}\mathcal{E}$ form a 2-category we denote by $\mathcal{L}\text{CAT}_{\mathcal{R}\mathcal{E}}$ at the top of the diamond of inclusion 2-functors below. We have biequivalences

$$\mathcal{L}\text{CAT}_{\mathcal{R}\mathcal{E}} \sim \text{CAT}_{\mathcal{R}*}(\mathcal{L}\text{Spn}\mathcal{E}) \sim \text{CAT}(\text{Spn}_{\mathcal{R}*}(\mathcal{L}\text{Spn}\mathcal{E})) .$$



Polynomials

Recall that polynomials in $\mathcal{E}$ are diagrams $p = (X \xleftarrow{r} A \xrightarrow{n} B \xrightarrow{t} Y)$ with n powerful. Our philosophy is that it is useful to regard p as a span

$$p = (X \xleftarrow{(n,A,r)} B \xrightarrow{(1_B,B,t)} Y)$$

in $\mathrm{Spn}\mathcal{E}$. Then the morphisms interpret as diagrams

$$\begin{array}{ccccc} & & B & & \\ & \swarrow (n,A,r) & \downarrow (1,B,g) & \searrow (1,B,t) & \\ X & \xleftarrow{(n',A',r')} & B' & \xrightarrow{(1,B',t')} & Y \end{array} \quad (1)$$

We modify this a bit by taking a compatible pair $(\mathcal{L}, \mathcal{R})$ of protocolibrations on $\mathcal{E}$ and restricting the polynomials p to those with $n \in \mathcal{L}$ and $t \in \mathcal{R}$. Restricting the morphisms of $\mathrm{Ply}\mathcal{E}$ in this way, we obtain a sub-bicategory which we denote by ${}_{\mathcal{L}}\mathrm{Ply}_{\mathcal{R}}\mathcal{E}$.

Theorem

If $(\mathcal{L}, \mathcal{R})$ is a compatible pair of protocolibrations on the category $\mathcal{E}$ then

$$(\mathcal{L}\text{Ply}_{\mathcal{R}}\mathcal{E})^{\text{op}} = \text{Spn}_{\mathcal{R}*}(\mathcal{L}\text{Spn}\mathcal{E}) .$$

Proof.

The 2-morphisms of $\text{Spn}_{\mathcal{R}*}(\mathcal{L}\text{Spn}\mathcal{E})$ are isomorphism classes of morphisms as pictured below.

$$\begin{array}{ccccc}
 & & B & & \\
 & \swarrow (1, B, t) & \downarrow (1, B, g) & \searrow (n, A, r) & \\
 Y & & \cong \quad \Downarrow \quad \lambda & & X \\
 & \swarrow (1, B', t') & \downarrow & \searrow (n', A', r') & \\
 & & B' & &
 \end{array}$$

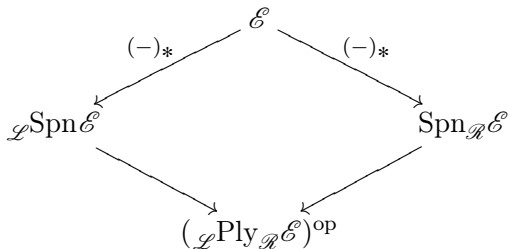
The data involved is exactly as in the diagram (1) in $\mathcal{E}$. Just note that the 2-morphism λ in $\mathcal{L}\text{Spn}\mathcal{E}$ is in the reverse direction as a morphism of spans.



Corollary

$$\mathrm{Ps}(\mathcal{L}\mathrm{Ply}_{\mathcal{R}\mathcal{E}}, \mathrm{Cat}) \sim \mathcal{L}\mathrm{CAT}_{\mathcal{R}\mathcal{E}}$$

$$\begin{aligned}\mathrm{Ps}(\mathcal{L}\mathrm{Ply}_{\mathcal{R}\mathcal{E}}, \mathrm{Cat}) &\sim \mathrm{CAT}(\mathrm{Spn}_{\mathcal{R}*}(\mathcal{L}\mathrm{Spn}\mathcal{E})) \\ &\sim \mathrm{CAT}_{\mathcal{R}*}(\mathcal{L}\mathrm{Spn}\mathcal{E}) \\ &\sim \mathcal{L}\mathrm{CAT}_{\mathcal{R}\mathcal{E}} .\end{aligned}$$



Suppose the bicategory $\mathcal{C}$ admits the binary bicomproduct

$$U \xrightarrow{i} U + V \xleftarrow{j} V$$

and $\mathcal{R}$ is a protocalibration of $\mathcal{C}$. For the pseudofunctor $\mathbb{C}_{\mathcal{R}} = \overrightarrow{\mathbf{1}} : \mathcal{C}^{\text{op}} \rightarrow \text{Cat}$, we have a canonical comparison functor

$$\mathbb{C}_{\mathcal{R}}(U + V) \rightarrow \mathbb{C}_{\mathcal{R}}U \times \mathbb{C}_{\mathcal{R}}V$$

taking $Z \xrightarrow{w} U + V$ to the pair $(X \xrightarrow{w_i} U, Y \xrightarrow{w_j} V)$ defined by bipullbacks

$$\begin{array}{ccc} X & \xrightarrow{i_w} & Z \\ w_i \downarrow \lrcorner & \xRightarrow{\cong} & \downarrow w \\ U & \xrightarrow{i} & U + V \end{array} \qquad \begin{array}{ccc} Y & \xrightarrow{j_w} & Z \\ w_j \downarrow \lrcorner & \xRightarrow{\cong} & \downarrow w \\ V & \xrightarrow{j} & U + V . \end{array}$$

Lemma

Suppose the bicategory $\mathcal{C}$ admits, and the protocolibration $\mathcal{R}$ is closed under, finite bicoproducts. Then the canonical comparison functors

$$\mathbb{C}_{\mathcal{R}}0 \rightarrow \mathbf{1} \quad \text{and} \quad \mathbb{C}_{\mathcal{R}}(U + V) \rightarrow \mathbb{C}_{\mathcal{R}}U \times \mathbb{C}_{\mathcal{R}}V$$

have left adjoints respectively defined on objects by

$$(0 \in \mathbf{1}) \mapsto (0 \xrightarrow{1_0} 0) \quad \text{and} \quad (X \xrightarrow{u \in \mathcal{R}} U, Y \xrightarrow{v \in \mathcal{R}} V) \mapsto (X + Y \xrightarrow{u+v \in \mathcal{R}} U + V).$$

Definition

Suppose $\mathcal{R}$ is a protocolibration of the bicategory $\mathcal{C}$. We say a bicategory $\mathcal{C}$ is $\mathcal{R}$ -extensive when it admits finite bicoproducts and these are taken into finite biproducts by the pseudofunctor $\mathbb{C}_{\mathcal{R}} : \mathcal{C}^{\text{op}} \rightarrow \text{Cat}$.

$$\mathbb{C}_{\mathcal{R}}0 \simeq \mathbf{1} \quad \text{and} \quad \mathbb{C}_{\mathcal{R}}(U + V) \simeq \mathbb{C}_{\mathcal{R}}U \times \mathbb{C}_{\mathcal{R}}V$$

Example

Suppose $\mathcal{E}$ is a lex extensive category. Then the maximum protocalibration $\mathcal{E}$ is closed under finite coproducts and $\mathcal{E}$ is $\mathcal{E}$ -extensive.

Also, the subcategory $\mathcal{P}$ consisting of the powerful morphisms is closed under finite coproducts, and $\mathcal{E}$ is $\mathcal{P}$ -extensive.

Proposition

Suppose the bicategory $\mathcal{C}$ is $\mathcal{R}$ -extensive and that the protocalibration $\mathcal{R}$ is closed under the finite bicoproducts. If $\mathbb{X} \in \text{CAT}\mathcal{C}$ preserves finite bicategorical products then the $\mathcal{R}$ -cocompletion $\overrightarrow{\mathbb{X}}$ also preserves finite bicategorical products.

Proposition

Suppose the bicategory $\mathcal{C}$ is $\mathcal{R}$ -extensive and that the protocalibration $\mathcal{R}$ is closed under the finite bicoproducts. Then

- i. the bicategory $\text{Spn}_{\mathcal{R}}\mathcal{C}$ admits finite bicoproducts, and*
- ii. each homcategory $\text{Spn}_{\mathcal{R}}\mathcal{C}(X, Y)$ admits finite coproducts preserved by composition on the diagrammatic right by any morphism $Y \xrightarrow{(u, S, v)} Z$ where $u \in \mathcal{R}$.*

For any bicategory $\mathcal{D}$ admitting finite bicoproducts, we write $\mathrm{fpCAT}\mathcal{D}$ for the full sub-bicategory of $\mathrm{CAT}\mathcal{D}$ consisting of the finite biproduct preserving pseudofunctors $\mathcal{D}^{\mathrm{op}} \rightarrow \mathrm{Cat}$. If $\mathcal{T}$ is a protocalibration of $\mathcal{D}$, we write $\mathrm{fpCAT}_{\mathcal{T}}\mathcal{D}$ for the intersection of $\mathrm{fpCAT}\mathcal{D}$ and $\mathrm{CAT}_{\mathcal{T}}\mathcal{D}$.

Proposition

Suppose the bicategory $\mathcal{C}$ is $\mathcal{R}$ -extensive and that the protocalibration $\mathcal{R}$ is closed under the finite bicoproducts. Then

$$\mathrm{fpCAT}(\mathrm{Spn}_{\mathcal{R}}\mathcal{C}) \sim \mathrm{fpCAT}_{\mathcal{R}}\mathcal{C} .$$

Example

With lexextensive $\mathcal{E}$, we now have that $\mathrm{Spn}\mathcal{E} = \mathrm{Spn}_{\mathcal{E}}\mathcal{E}$ has bicoproducts and local coproducts preserved by right composition. However, $\mathrm{Spn}\mathcal{E} = (\mathrm{Spn}\mathcal{E})^{\mathrm{op}}$, so local coproducts are preserved by composition on the left as well. Furthermore, if $\mathcal{R}$ is any protocalibration of the lexextensive category $\mathcal{E}$ closed under finite coproducts then $\mathcal{E}$ is $\mathcal{R}$ -extensive. The bicoproducts and local coproducts in $\mathrm{Spn}_{\mathcal{R}}\mathcal{E}$ are preserved by the inclusion pseudofunctor into $\mathrm{Spn}\mathcal{E}$. It follows that the local coproducts in $\mathrm{Spn}_{\mathcal{R}}\mathcal{E}$ are preserved by composition on both sides. Consider the symmetric closed monoidal bicategory Cat_+ of categories with finite coproducts and finite-coproduct-preserving functors; the internal hom $[\mathcal{A}, \mathcal{B}]_+$ of Cat_+ is $\mathrm{Cat}_+(\mathcal{A}, \mathcal{B})$ with pointwise coproduct and the unit object is the category set of finite sets. Cartesian product $A \times B$ of categories is a direct sum in Cat_+ and the terminal category $\mathbf{1}$ is also initial. What we have is that $\mathrm{Spn}\mathcal{E}$ is Cat_+ -enriched and to say a pseudofunctor $(\mathrm{Spn}\mathcal{E})^{\mathrm{op}} \rightarrow \mathrm{Cat}_+$ is biproduct preserving is to say it is Cat_+ -enriched.

Lemma

Suppose $(\mathcal{L}, \mathcal{R})$ is a compatible pair of protocolibrations on an $\mathcal{R}$ -extensive category $\mathcal{E}$. Suppose both $\mathcal{L}$ and $\mathcal{R}$ are closed under finite coproducts in $\mathcal{E}$. The protocolibration $\mathcal{R}_$ of the bicategory $\mathcal{L}\mathrm{Spn}\mathcal{E}$ (see Lemma 6) is closed under bicoproducts and $\mathcal{L}\mathrm{Spn}\mathcal{E}$ is $\mathcal{R}_*$ -extensive.*

Theorem

Under the hypotheses of the Lemma, the bicategory $\mathcal{L}\mathrm{Ply}_{\mathcal{R}}\mathcal{E}$ admits finite biproducts and its homs admit finite coproducts preserved by composition on the diagrammatic left.

Objective Mackey functors

Suppose the bicategory $\mathcal{C}$ is finitely bicocomplete and $\mathcal{R}$ -extensive for a protocalibration $\mathcal{R}$ which, when regarded as a wide subcategory of $\mathcal{C}$, is closed under finite bicoproducts.

Definition

An *objective $\mathcal{R}$ -Mackey functor* on $\mathcal{C}$ is an $\mathcal{R}$ -cocomplete pseudofunctor $\mathbb{M} : \mathcal{C}^{\text{op}} \rightarrow \text{Cat}$ which preserves finite biproducts (in the bicategorical sense). We define the bicategory of objective $\mathcal{R}$ -Mackey functors by $\text{OMky}_{\mathcal{R}}\mathcal{C} = \text{fpCAT}_{\mathcal{R}}\mathcal{C}$. We have:

$$\text{fpCAT}(\text{Spn}_{\mathcal{R}}\mathcal{C}) \sim \text{OMky}_{\mathcal{R}}\mathcal{C} .$$

When $\mathcal{R} = \mathcal{C}$, we drop the subscripts $\mathcal{C}$ and refer to *objective Mackey functors*.

Example

The object $\mathbb{C}_{\mathcal{R}} \in \text{CAT}_{\mathcal{R}}\mathcal{C}$ is an objective $\mathcal{R}$ -Mackey functor and is called the *objective $\mathcal{R}$ -Burnside functor*.

Example

If $\mathbb{Z}$ is an objective $\mathcal{R}$ -Mackey functor then so is $[\mathbb{Y}, \mathbb{Z}]_{\mathcal{R}}$ where

$$[\mathbb{Y}, \mathbb{Z}]_{\mathcal{R}} W \simeq \text{CAT}_{\mathcal{R}}\mathcal{C}(\mathbb{Y}, \mathbb{Z}(- \times W)) .$$

Proposition

Every objective $\mathcal{R}$ -Mackey functor $\mathbb{M} : \mathcal{C}^{\text{op}} \rightarrow \text{Cat}$ factors through the forgetful 2-functor $\text{Cat}_+ \rightarrow \text{Cat}$ by a pseudofunctor also denoted by $\mathbb{M}$.

Proof.

By assumption, the codiagonal $\nabla : U + U \rightarrow U$ and unique $! : 0 \rightarrow U$ are in $\mathcal{R}$. The diagonal of $\mathbb{M}U$ is isomorphic to the composite

$\mathbb{M}U \xrightarrow{\mathbb{M}\nabla} \mathbb{M}(U + U) \simeq \mathbb{M}U \times \mathbb{M}U$ and so has a left adjoint. Thus $\mathbb{M}U$ has binary coproducts. Similarly, $\mathbb{M}U \xrightarrow{\mathbb{M}!} \mathbb{M}0 \simeq \mathbf{1}$ has a left adjoint so $\mathbb{M}U$ has an initial object. For $f : U \rightarrow V$, the functor $\mathbb{M}f$ preserves these finite coproducts; apply the CB property to these pullbacks in our $\mathcal{R}$ -extensive bicategory $\mathcal{C}$.

$$\begin{array}{ccc} U + U & \xrightarrow{\nabla} & U \\ f+f \downarrow & & \downarrow f \\ V + V & \xrightarrow{\nabla} & V \end{array},$$

$$\begin{array}{ccc} 0 & \xrightarrow{!} & U \\ 1_0 \downarrow & & \downarrow f \\ 0 & \xrightarrow{!} & V \end{array}.$$

Corollary

If $\mathbb{M}$ is an objective $\mathcal{R}$ -Mackey functor and $U \xrightarrow{i} U + V \xleftarrow{j} V$ is a coproduct in $\mathcal{C}$ then

$$\begin{array}{ccc} \mathbb{M}U & \xrightleftharpoons[\mathbb{M}i]{\mathbb{M}_*i} & \mathbb{M}(U + V) \xrightleftharpoons[\mathbb{M}_*j]{\mathbb{M}j} \mathbb{M}V \\ & \perp & \top \end{array}$$

exhibits a (bicategorical) direct sum in Cat_+ .

Corollary

For all $f : X' \rightarrow X$ in $\mathcal{C}$, the functor $\mathbb{C}_{\mathcal{R}}f : \mathbb{C}_{\mathcal{R}}X \rightarrow \mathbb{C}_{\mathcal{R}}X'$ preserves finite coproducts and the pseudofunctor

$$\mathcal{C}^{\text{op}} \times \mathcal{C} \xrightarrow{(-)_* \times (-)_*} (\text{Spn}_{\mathcal{R}})^{\text{op}} \times \text{Spn}_{\mathcal{R}} \xrightarrow{\text{Spn}_{\mathcal{R}}(-, -)} \text{Cat}$$

factors through the inclusion $\text{Cat}_+ \hookrightarrow \text{Cat}$ of the sub-2-category of categories with finite coproducts and functors which preserve them.

Remark

We spoke of a hom enrichment of $\mathrm{Spn}_{\mathcal{R}}\mathcal{C}$ in Cat_+ in the ordinary sense. For $\mathrm{Spn}_{\mathcal{R}}\mathcal{C}$, we have an example of enrichment in a skew base which is an objective version of a base observed in 2012 by Cockett and Lack in connection with left additive (= left linear over the ring of integers) categories; see page 1101 of Garner-Lemay (2021). What we notice is that the sub-2-category Cat_+ of Cat has a skew closed structure with internal hom of $A, B \in \mathrm{Cat}_+$ taken as the ordinary functor category $[\mathcal{A}, \mathcal{B}]$ where the finite coproducts are pointwise; write $\mathrm{Cat}_{+\mathrm{sk}}$ for Cat_+ equipped with this skew-monoidal structure. Then $\mathrm{Spn}_{\mathcal{R}}\mathcal{C}$ is a skew $\mathrm{Cat}_{+\mathrm{sk}}$ -enrichment of $\mathcal{C}$ in the sense of Campbell where the enriched hom is $\mathrm{Spn}_{\mathcal{R}}\mathcal{C}(X, Y)$. In particular, we have that $\mathcal{L}\mathrm{Ply}_{\mathcal{R}}\mathcal{C}$ is a skew enrichment of $\mathcal{L}\mathrm{Spn}\mathcal{C}$ in $\mathrm{Cat}_{+\mathrm{sk}}$.

Proposition

The $\mathcal{R}$ -cocompletion pseudomonad $\overrightarrow{(-)}$ on $\text{CAT}\mathcal{C}$ restricts to a pseudomonad on the bicategory $\text{fpCAT}\mathcal{C}$ of finite biproduct preserving pseudofunctors $\mathcal{C}^{\text{op}} \rightarrow \text{Cat}$. The Eilenberg-Moore pseudo-algebras are the objective $\mathcal{R}$ -Mackey functors.

There is a symmetric monoidal closed structure on $\text{OMky}_{\mathcal{R}\mathcal{C}}$.

$$\text{OMky}_{\mathcal{R}\mathcal{C}}(\mathbb{M} \star \mathbb{N}, \mathbb{L}) \simeq \text{OMky}_{\mathcal{R}\mathcal{C}}(\mathbb{M}, [\mathbb{N}, \mathbb{L}]_{\mathcal{R}}) .$$

The unit for this tensor product is the objective $\mathcal{R}$ -Burnside functor $\mathbb{C}_{\mathcal{R}}$. The monoidales (pseudomonoids) for the convolution monoidal structure $\mathbb{A} \star \mathbb{B}$ we call *objective $\mathcal{R}$ -Green functors on $\mathcal{C}$* and write $\text{OGrn}_{\mathcal{R}\mathcal{C}}$ for the bicategory they, with the strong monoidal morphisms, form. An objective $\mathcal{R}$ -Green functor is *symmetric* when it is symmetric as a monoidale.

An $\mathcal{R}$ -Mackey functor on $\mathcal{E}$ is a $\mathbf{CMon}$ -enriched (or equally, a finite-product-preserving) functor

$$M : (\mathrm{clSpn}_{\mathcal{R}}\mathcal{E})^{\mathrm{op}} \rightarrow \mathbf{CMon} .$$

This is consistent with Lindner (1976) who looked at the case $\mathcal{R} = \mathcal{E}$. Put

$$\mathrm{Mky}_{\mathcal{R}}\mathcal{E} = [(\mathrm{clSpn}_{\mathcal{R}}\mathcal{E})^{\mathrm{op}}, \mathbf{CMon}]_+ ,$$

the $\mathbf{CMon}$ -enriched functor category. It is a monoidal $\mathbf{CMon}$ -enriched category by Day convolution. The monoids for this convolution structure are $\mathcal{R}$ -Green functors on $\mathcal{E}$. The Burnside functor $E_{\mathcal{R}}$ is an example.

Let $\mathbf{RLxt}$ denote the 2-category whose objects are pairs $(\mathcal{A}, \mathcal{O})$ consisting of a lexextensive category $\mathcal{A}$ and a protocalibration $\mathcal{O}$ closed under finite coproducts, and, whose morphisms $F : (\mathcal{A}, \mathcal{O}) \rightarrow (\mathcal{A}', \mathcal{O}')$ are functors $F : \mathcal{A} \rightarrow \mathcal{A}'$ preserving pullback and finite coproducts, and taking $\mathcal{O}$ to $\mathcal{O}'$. Then we have a pseudofunctor

$$\mathbf{Mky}_R : \mathbf{RLxt} \rightarrow \mathbf{Cat}_+$$

taking $(\mathcal{A}, \mathcal{O})$ to $\mathbf{Mky}_{\mathcal{O}} \mathcal{A}$ regarded as a category with finite coproducts.

It is useful to notice that the objective Burnside functor $\mathbb{E}_R : \mathcal{E}^{\text{op}} \rightarrow \mathbf{Cat}_+$ lands in the category $\mathbf{RLxt}$ over $\mathbf{Cat}_+$ by equipping each lexextensive category $\mathbb{E}_R(X) (\subseteq \mathcal{E}_{/X})$ with the protocalibration $\mathcal{R}_{/X}$.

Objective Green functor of abstract Mackey functors

Proposition

The composite pseudofunctor

$$\mathrm{Mky}_{\mathbf{R}\mathcal{R}} := \mathcal{C}^{\mathrm{op}} \xrightarrow{\mathbb{E}_{\mathcal{R}}} \mathbf{RLxt} \xrightarrow{\mathrm{Mky}_{\mathbf{R}}} \mathbf{Cat}_+$$

becomes a symmetric objective $\mathcal{R}$ -Green functor when equipped with the multiplication induced by the finite-coproduct-preserving-in-each-variable functors

$$\mathrm{Mky}_{\mathbf{R}\mathcal{R}}\mathbb{E}_{\mathcal{R}}X \times \mathrm{Mky}_{\mathbf{R}\mathcal{R}}\mathbb{E}_{\mathcal{R}}Y \rightarrow \mathrm{Mky}_{\mathbf{R}\mathcal{R}}\mathbb{E}_{\mathcal{R}}(X \times Y)$$

taking (M, N) to $M \boxtimes N$ where

$$(M \boxtimes N)(S \xrightarrow{(a,b)} X \times Y) = M(S \xrightarrow{a} X) \otimes N(S \xrightarrow{b} Y) .$$

Objective Tambara functors

Suppose $\mathcal{E}$ is finitely complete and the pair $(\mathcal{L}, \mathcal{R})$ satisfies the hypotheses of the Lemma on page 20.

Definition

An *objective $(\mathcal{L}, \mathcal{R})$ -Tambara functor* on $\mathcal{E}$ is a finite biproduct preserving pseudofunctor $\mathbb{T} : \mathcal{E}^{\text{op}} \rightarrow \text{Cat}$ which is in ${}_{\mathcal{L}}\text{CAT}_{\mathcal{R}}^{\mathcal{E}}$. We define the bicategory of objective $(\mathcal{L}, \mathcal{R})$ -Tambara functors by

$${}_{\mathcal{L}}\text{Tmb}_{\mathcal{R}}^{\mathcal{E}} := \text{fp } {}_{\mathcal{L}}\text{CAT}_{\mathcal{R}}^{\mathcal{E}} .$$

$$\text{fpCAT}(({}_{\mathcal{L}}\text{Ply}_{\mathcal{R}}^{\mathcal{E}})^{\text{op}}) \sim {}_{\mathcal{L}}\text{Tmb}_{\mathcal{R}}^{\mathcal{E}} .$$

Example

If $\mathbb{Z}$ is a $(\mathcal{L}, \mathcal{R})$ -Tambara functor then so is $[\mathbb{Y}, \mathbb{Z}]_{\mathcal{R}}$ for all $\mathbb{Y} \in \text{CAT}_{\mathcal{R}\mathcal{E}}$.

Example

Take $\mathcal{E}$ to be a cartesian closed category cat of small categories and functors. Let $\mathcal{X}$ be a complete and cocomplete category. Then $[-, \mathcal{X}] : \text{cat}^{\text{op}} \rightarrow \text{Cat}$ is a $(\mathcal{P}, \mathcal{E})$ -Tambara functor.

Lemma

$\mathcal{E}$ is $\mathcal{L}$ -extensive.

Corollary

If $\mathbb{X} \in \text{CAT}$ preserves biproducts then so do $\overleftarrow{\mathbb{X}}$ and $\overrightarrow{\mathbb{X}}$.

Proposition

The composite pseudomonad $\overleftrightarrow{(-)}$ on $\text{CAT}^{\mathcal{E}}$ restricts to a pseudomonad on the bicategory $\text{fpCAT}^{\mathcal{E}}$ of finite product preserving pseudofunctors $\mathcal{E}^{\text{op}} \rightarrow \text{Cat}$. The pseudo-algebras are the objective $(\mathcal{L}, \mathcal{R})$ -Tambara functors.

An $(\mathcal{L}, \mathcal{R})$ -Tambara functor on $\mathcal{E}$ is a finite-product-preserving functor

$$T : \text{cl}_{\mathcal{L}} \text{Ply}_{\mathcal{R}} \mathcal{E} \rightarrow \text{CMon}.$$

This concept, for the case $\mathcal{L} = \mathcal{R} = \mathcal{E} = G\text{-set}$, arose in Tambara (1993); a version of $\mathcal{L}$ and $\mathcal{R}$ came later, called *bi-incomplete Tambara functors*.

Thank You



$$\text{M} \xrightarrow[\Delta]{\text{CoACT}} \text{MQ}$$